BIOT 6900 · AI-POWERED BIOTECH WORKFLOWS

Module 6 Lab Guide, Part 2: Applying a Real PRS to Real Genotypes

Week 6 · the real case · instructor-led, not graded · Fall 2026

Notebook: `BIOT6900_Module6_Part2_RealPRS.ipynb` Runtime: about 20 minutes Real data from Canvas, or a synthetic fallback

What this lab is

In Part 1 you built a PRS by hand on invented data. Here you run the same workflow the way a real analyst would: take a published Alzheimer's score from the PGS Catalog, apply it to real 1000 Genomes genotypes, and ask the one question the toy model could not, does the score work equally well across ancestry groups?

It is worked, not graded. We run each cell together and read the output. You do the whole thing yourself, graded, in Part 3.

Step	What you run	The real thing it is
1	Load real genotypes, a real score, and ancestry labels	1000 Genomes + PGS Catalog
2	Harmonize the effect allele and compute the PRS	plink --score, an applied PGS
3	Plot the PRS distribution per ancestry	The portability / fairness check
4	AUC overall and within each ancestry	Discrimination validation
5	Calibration curve	Are the probabilities honest
6	PRS on top of age	Incremental clinical value

The one idea to carry through

The arithmetic is identical to Part 1. What changes on real data is that the score no longer works the same for everyone, and this analysis shows you exactly for whom it fails.

Maps to course outcomes: LO1 (apply industry-standard tools), LO3 (interpret results in clinical context), LO4 (evaluate model performance, limitations, and validation).

Before you start

1 · Your environment

The biot6900 environment from Module 1, which already has numpy, pandas, matplotlib and scikit-learn. Nothing to install.

2 · The data

The notebook looks for four files in a `data/` folder posted on Canvas: `genotypes.csv` (a 1000 Genomes dosage subset; dosages count the ALT allele), `variants.csv` (variant, ref, alt), `score.csv` (the PGS Catalog file: variant, effect_allele, other_allele, weight), and `meta.csv` (ancestry, age, phenotype). If any are absent it uses a clearly-labelled synthetic demo cohort, so the lab always runs.

What is real, and what is simulated

The genotypes, the published weights, the ancestry labels, and the PRS and its distribution across groups are real. Only the case/control label is simulated, because 1000 Genomes has no disease labels. The AUC and calibration numbers are therefore illustrative, but the portability pattern they reveal is real.

The real case

We use Alzheimer's disease, to connect to the APOE target from Part I, and a PGS Catalog score trained in European-ancestry samples, like most published scores. The cohort is split into 1000 Genomes super-populations: EUR, AFR, EAS, SAS and AMR. The European training ancestry is the root of what you are about to see.

Do this before running anything

Predict: in which ancestry group will a European-trained score discriminate best, and will the gap to the others be small or large? Write your answer down. Step 4 checks it.

Walking through the steps

Step 1 · Load the data

What it does. Loads the genotypes, the score, and the ancestry/age/phenotype table, or falls back to the synthetic demo with a loud warning.

What you should see. A line reporting the number of people, the number of score variants, and the case rate (about 30%).

The point. Real analysis begins with bookkeeping. Know exactly what you loaded before computing anything.

Step 2 · The score we apply

What it does. Shows the scoring file: each variant with an effect allele and a per-allele log-odds weight.

What you should see. A table of variants and weights, trained in European samples.

The point. A PGS is just a weighted variant list. Nothing about it is tailored to the person you apply it to, which is why portability is not automatic.

Step 3 · Harmonize and compute the PRS

What it does. For each variant it compares the score's effect allele with the REF and ALT nucleotides: a match to ALT is used as is, a match to REF is flipped with `2 - dosage`, and a match to neither is dropped and reported. Then it computes `prs = G @ w`.

What you should see. A line reporting variants used, flipped, and dropped, then a standardized PRS for everyone.

The point. Effect-allele alignment is the single most common PRS bug. A variant matching neither allele (strand or build mismatch) is dropped, not guessed. The score itself is the same one line as Part 1.

Step 4 · The distribution shifts across ancestries

What it does. Plots the PRS distribution for each ancestry group and prints the mean PRS per group.

What you should see. The groups centred at different places, and a printed table of mean PRS and the fraction of each group above a single top-10% cutoff: for example about 24% of one group versus about 4% of another. This panel uses no phenotype, only real genotypes and real weights.

The point. A single fixed percentile cutoff would flag very different fractions of each group. That is a fairness problem before any disease label enters.

Step 5 · Discrimination by ancestry

What it does. Computes the AUC overall and within each ancestry, using the simulated label.

What you should see. AUC highest in EUR (about 0.68) and lower elsewhere (about 0.53 to 0.57). Check it against your prediction.

The point. Same score, same biology, worse performance outside the training ancestry, because the European weights tag the causal variants less well where LD and allele frequencies differ. This is the portability problem, quantified.

Step 6 · Calibration by ancestry

What it does. Fits the risk model in the EUR group only, applies it to everyone, and plots one calibration curve per ancestry.

What you should see. EUR sitting on the diagonal, and the other groups drifting off it: predicted risk no longer matches observed risk.

The point. A model calibrated in one ancestry is less honest when applied to another. Fitting and testing on the same pooled data could never reveal this, which is why the model is fit in EUR and tested everywhere.

Step 7 · Does the PRS beat age alone?

What it does. Compares cross-validated AUC for age only versus age plus the PRS.

What you should see. A small rise, roughly 0.54 to 0.62.

The point. The incremental value is real but modest, the honest expectation for a common-disease PRS. It is one input among many, not a verdict.

What maps to what

This step	The real practice	What production adds	What it does not fix
load real files	1000 Genomes + PGS Catalog	QC, imputation, build matching	cohorts are still mostly European
G @ w	plink --score, applied PGS	missing-genotype handling at scale	still a weighted sum
AUC by ancestry	validation across cohorts	real labels, confidence intervals	the portability gap itself
calibration	recalibration per population	per-ancestry intercept/slope fits	needs labelled data per group

Troubleshooting

Symptom	Likely cause and fix
“NO real data in ./data” warning	Expected if the four Canvas files (including variants.csv) are not all in data/. The synthetic demo runs and teaches the same workflow. Add the files to see real numbers.
Very few variants after merging	Variant IDs or effect alleles do not match. Harmonize IDs and align the effect allele before summing.
Cases score lower than controls	Effect allele not aligned. Re-check the keep/flip/drop rule in Step 3 against REF and ALT.
Many variants dropped at harmonization	The effect allele matches neither REF nor ALT: usually a strand or genome-build mismatch between genotypes.csv and the score. A few drops are normal; a large count needs investigating.
AUC does not drop across ancestries	You standardized the PRS within each group, which hides the shift. Standardize once, using a single reference.
Numbers differ from this guide	Run Kernel → Restart & Run All. Cells depend on earlier ones.

What this lab deliberately leaves out

- Real disease labels. The phenotype is simulated, so AUC and calibration are illustrative, not clinical.
- One disease, one score. You see a single European-trained score; recalibration and cross-ancestry methods are not applied here.
- Covariates beyond age, and any treatment of related individuals or population structure correction.

What comes next

Part 3, the graded assignment, has you run this entire workflow yourself on a PGS Catalog score for your own Part I disease, and write up the portability, calibration, and clinical-utility questions. The reading habits from this lab transfer directly.